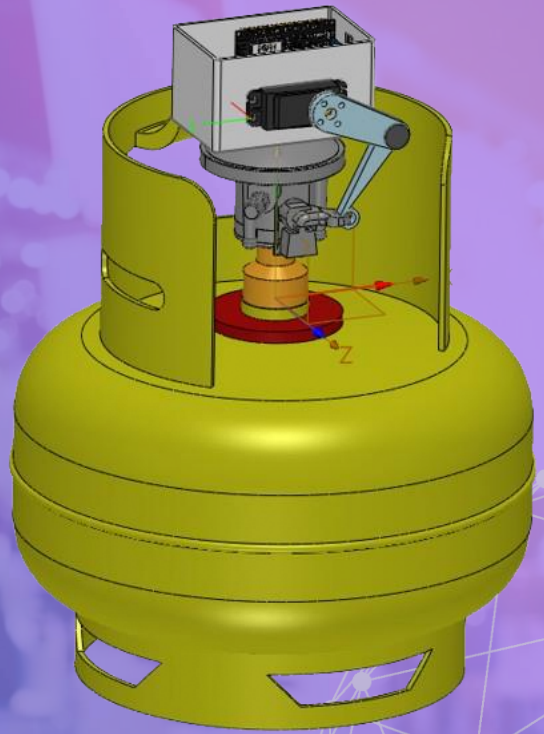
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SIGAS

Smart Integrated Gas Security: Autonomous LPG Leak Mitigation & Multi-Unit IoT Valve Control for Culinary Micro-Enterprises



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Problem Statement

In today's food industry, micro-culinary businesses (UMKM Kuliner) rely heavily on 3 kg LPG cylinders for 12 to 16 continuous hours daily in cramped kitchen kiosks. But unfortunately, safety protocols in traditional food centers are rarely adequate to prevent catastrophic gas leaks.

Gas leaks and fires pose severe threats to grassroots vendors. In our field interviews in Purwokerto Selatan, 100% of respondents experienced gas smells, loose seals, or damaged hoses—risking up to IDR 19M (~$1,200 USD) in asset loss per kiosk, excluding damage to neighboring stalls.

What's more concerning is that vendors routinely leave regulators clamped after closing hours. When stalls shut down for the night, operators turn off the stove knobs but leave regulators pressurized. If a micro-crack develops overnight, flammable gas silently accumulates inside the locked kiosk.

On the other hand, conventional gas and smoke detectors fail to address the core danger. They merely emit a high-pitched buzzing alarm to an empty room, leaving the gas flowing continuously until a single spark ignites an uncontrollable fire.

Target User & Market

1. Individual Users (Culinary MSMEs)

- **Segments**: Street food vendors, warung makan, noodle kiosks, night market food stalls.
- **Needs:** Affordable fire protection, zero pipe alteration, automatic valve cut-off during unattended night hours.

2. SMEs and Digital Startups

- **Segments:** Food court managers, culinary centers (sentra UMKM), cloud kitchens, vocational school canteens.
- **Needs:** Centralized digital safety supervision across 10–50 rented stalls, compliance enforcement, and fire liability prevention.

3. Geographic Focus

- **Primary:** Indonesia (over 4.5 million culinary MSMEs relying on subsidized 3 kg LPG cylinders).
- **Expansion Potential:** Southeast Asia (similar dense street-food culture and identical gas cylinder safety challenges).

4. Demographics

- **Age**: 20-55 years old
- **Profile:** Food stall owners, cook assistants, and commercial food court administrators managing shared utility spaces.

Proposed Solution

To address the severe fire hazards and unattended gas leaks in culinary MSMEs, our proposed solution is to develop SIGAS (Smart IoT Gas Autonomous Safety)—a non-invasive, edge-autonomous robotic cut-off actuator paired with a centralized multi-unit mobile monitoring application that physically shuts off LPG regulators before fires can ignite, without requiring gas pipe modifications or technical expertise..

End goal, empower users to:

- Take proactive control of their gas safety through automated daily off-hours regulator cut-offs.
- Achieve immediate mechanical shut-off within 15 seconds of gas detection, even during complete power or internet outages.
- Centrally supervise multi-kiosk gas safety and operational compliance in real time through an intuitive mobile dashboard.

Key Features & Tech Stack

- **Microcontroller**
  ESP32 DOIT DevKit V1 (240MHz Dual-Core, Wi-Fi/BLE).
- **Gas Sensing**
  Calibrated MQ-6 LPG semiconductor sensor with 10-sample Moving Average Filter (10 Hz sampling rate).
- **Actuation**
  TD8120MG 20 kg•cm metal-gear waterproof servo motor for rapid 90° valve shutoff.
- **Timekeeper**
  DS3231 I2C Real-Time Clock with battery backup (±2 ppm accuracy).
- **Acoustic Alert**
  12V active high-decibel piezo buzzer (distinct pairing and emergency audio patterns).
- **Mobile Framework**
  Flutter 3 (Dart) for smooth 60fps Android user experience.
- **Cloud Backend**
  Firebase Realtime Database & Cloud Messaging for sub-second telemetry.
- **Edge Persistence**
  Non-Volatile Storage (NVS/Preferences) for localized shift timers.
- **Zero-Ghost Security**
  QR code pairing scanner with live hardware online ping verification.
- **Code Reliability**
  Non-blocking state machine with separate emergency interlock and manual controls.

Screenshots or Product Demo

SIGAS Kitchen Safety Apps

This integrated IoT ecosystem is designed to help micro-culinary vendors and food center managers protect their physical assets and lives with ease. It combines a non-invasive robotic servo actuator that clamps onto existing LPG regulators with a sleek mobile dashboard featuring 270° radial gas monitoring, automated operational shift scheduling, and confirmed remote valve override controls. Whether you are a solo warung owner or managing a bustling 30-stall food court, SIGAS empowers you to maintain complete kitchen fire safety—without requiring any plumbing modifications or technical expertise



1. Normal Operation



2. Early Leak Warning



3. Emergency Auto-Cutoff

Competitive Advantage

1. Non-Invasive Plug-On Design, Clamps directly onto standard LPG regulators without cutting gas hoses, preserving safety certifications with instant manual hand release.

2. Edge-Autonomous Safety, Built-in ESP32 and RTC clock execute physical shut-offs in under 15 seconds—even during complete power or internet blackouts.

3. Multi-Unit Cluster Supervision, Food court managers can monitor and protect 10–50 kitchen stalls simultaneously from a single centralized mobile dashboard.

4. Affordable Protection, Delivers automated robotic valve cut-off at IDR 850K (~$55 USD)—less than one-tenth the cost of expensive industrial systems.

Current Status & Development Plan

Current Status (TRL 6 Prototype)

- **Working Prototype**: ESP32, MQ-6 sensor, and 20kg servo mounted on 3D plug-on casing.
- **Smart Firmware:** 10Hz moving average filter, RTC scheduling, and NVS flash memory.
- **Mobile App:** Flutter 3 dashboard with real-time Firebase syncing and QR pairing.
- **Field Validated:** Surveyed and verified with 3 culinary MSMEs in Purwokerto

Current Status & Development Plan

Testing & Pilot Plan (4–8 Weeks)

- 500-cycle mechanical endurance test on brass regulator levers
- Deploy 5 pilot units across vocational school canteens
- Transition 3D casing to flame-retardant injection-molded ABS/PC

Deliverables:

- Mechanical endurance & torque calibration logs
- Beta feedback summary from canteen vendors
- Safety audit & certification checklist

Business Model

Revenue Streams

• Hardware Direct Unit Sales:

Direct B2B sales of SIGAS kits (MSRP IDR 850,000 / ~$55 USD; direct BOM IDR 549,780; healthy 35.3% contribution margin).

• Safety-as-a-Service (SaaS Subscription):

IDR 25,000/stall/month for multi-unit food court management dashboard, SMS emergency alerts, and quarterly calibration.

• Replacement Accessories & Adapters:

Modular 3D mounts tailored for various regional commercial regulator geometries.

Distribution Channels

• Direct outreach to culinary centers, food courts, and market cooperatives (Koperasi Pedagang).

• Partnership programs with municipal vocational education & MSME agencies (Dinas Koperasi & UMKM).

• Online B2B platforms and authorized LPG equipment distributors.

Impact Plan

Improve Personal Data Protection Behavior

Goal: Protect vulnerable food micro-entrepreneurs from catastrophic fire disasters by automating mechanical regulator cut-offs and establishing centralized safety monitoring.

Target Impact:

- 0% unattended off-hours kitchen fires across all deployed food court clusters and canteens.
- 100% mechanical shut-off reliability within 15 seconds during gas leak incidents without user intervention.
- Protect over IDR 12.5M (~$800 USD) in vulnerable vendor cooking assets and carts per protected stall.
- Prevent catastrophic fire contagion from spreading across densely packed commercial market kiosks.

Roadmap & Future Direction

Here's the roadmap of developing the apps

0-6 Months
Goal: Validate 500-cycle valve mechanical endurance, deploy 10 pilot units in Purwokerto canteens, and secure national safety patents

6-12 Months
Goal: Launch the B2B web management portal, partner with local trade and MSME agencies, and produce flame-retardant injection-molded casings

12-18 Months
Goal: Scale user base through B2B institutional deals with municipal food courts and integrate NB-IoT for off-grid stalls.

18-24 Months
Goal: Build trust across 5 major culinary metro hubs in Java, grow recurring SaaS revenue, and create a certified safety ecosystem.

2+ Years
Goal: Adapt modular mounts for Vietnam, Thailand, and Philippines, becoming the leading culinary IoT safety platform in Southeast Asia.

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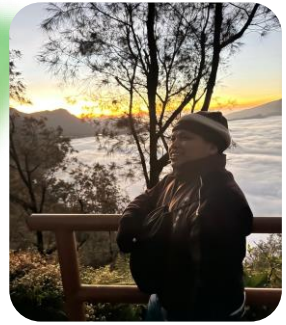
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